



Yoonhyuk Choi (Last update: Jun. '26)

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Sookmyung Women's University
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RESEARCH INTEREST

My research interests lie in **Machine Learning** and **Data Mining**, particularly in *Large Language Models*, *Graph Neural Networks*, and *Recommender Systems*.

EDUCATION

- **Seoul National University** Mar. 2019 - Aug. 2023
Ph.D. & M.S., Computer Engineering
Advisor: [Chong-Kwon Kim](#)
- **University of Seoul** Mar. 2013 - Feb. 2019
B.S., Computer Science
Advisor: [Eui-Kyeong Hong](#)

WORK EXPERIENCE

- **Assistant Professor** Sep. 2025 -
Sookmyung Women's University, Seoul, South Korea
Department of AI
- **Research scientist** Oct. 2024 - Aug. 2025
Samsung SDS, Seoul, South Korea
Preceding AI Lab (LLM, RAG)
- **Postdoc. associate** Nov. 2023 - Sep. 2024
Arizona State University, Tempe, United States (PI: [Selcuk Candan](#) and [Huan Liu](#))
School of Computing and Augmented Intelligence (SCAI)
- **Postdoc. associate** Sep. 2023 - Nov. 2023
Korea Institute of Energy Technology, Naju, South Korea (PI: [Chong-Kwon Kim](#))
Energy AI
- **Backend engineer** Jun. 2018 - Sep. 2018
nTOPAZ, Seoul, South Korea
Back-end Engineer

PUBLICATIONS ([GOOGLE SCHOLAR](#)) - C: CONFERENCE, J: JOURNAL

- (C20) Gauge-Equivariant Graph Networks via Self-Interference Cancellation ([link](#))
Yoonhyuk Choi, Jiho Choi, Jiwoo Kang
ICML '26
- (C19) Delay-Aware Sequential Recommendation with Dynamic Graphs and Variate-Temporal Decomposition ([link](#))
Yoonhyuk Choi
SIGIR '26 (short)

- (C18) Sheaf Graph Neural Networks via PAC-Bayes Spectral Optimization ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Jongwook Kim, Chong-Kwon Kim
AAAI '26
- (J17) Identifying Heterophilic Neighbors via Confidence-based Subgraph Matching for Graph Neural Networks ([link](#))
Yoonhyuk Choi, Chong-Kwon Kim
Artificial Intelligence '26 (IF: 4.6)
- (J16) Beyond Binary: Improving Signed Message Passing in Graph Neural Networks for Multi-Class Graphs ([link](#))
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim
IEEE TPAMI '25 (IF: 20.8)
- (J15) Hierarchical Hyperbolic Embeddings for Review-Driven Cross-Domain Recommendation ([link](#))
Yoonhyuk Choi, Chong-Kwon Kim
IEEE Access '25 (IF: 3.6)
- (J14) Generalization of Knowledge Transfer with User Reviews for Cross-Domain Recommendation ([link](#))
Yoonhyuk Choi, Chong-Kwon Kim
IEEE Access '25 (IF: 3.6)
- (C13) Selective Blocking for Message-Passing Neural Networks on Heterophilic Graphs ([link](#))
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim
UAI '25
- (C12) Review-Based Hyperbolic Cross-Domain Recommendation ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim
WSDM '25
- (C11) Mitigating Overfitting in Graph Neural Networks via Feature and Hyperplane Perturbation ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim
WSDM '25
- (J10) Beyond Message-Passing: Generalization of Graph Neural Networks via Feature Perturbation for ... ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim
IEEE TNNLS '24 (IF: 14.25)
- (C9) Improving the Text Convolution Mechanism with Large Language Model for ... ([link](#))
Yoonhyuk Choi, Fahim Tasneema Azad
IEEE Big Data '24 (short)
- (C8) Prioritizing Potential Wetland Areas via Region-to-Region Knowledge Transfer and Adaptive Propagation ([link](#))
Yoonhyuk Choi, Reepal Shah, John Sabo, Selcuk Candan, Huan Liu
IEEE Big Data '24
- (C7) Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning ([link](#))
Ahmet Kapkic, Pratanu Mandal, Shu Wan, Paras Sheth, Abhinav Gorantla, Yoonhyuk Choi, Huan Liu, K Selçuk Candan
CIKM '24 (benchmark)
- (C6) Universal Graph Contrastive Learning with a Novel Laplacian Perturbation ([link](#))
Taewook Ko, Yoonhyuk Choi, Chong-Kwon Kim
UAI '23
- (J5) A spectral graph convolution for signed directed graphs via magnetic laplacian ([link](#))
Taewook Ko, Yoonhyuk Choi, Chong-Kwon Kim
Neural Networks '23 (IF: 7.8)
- (J4) Aspect-oriented unsupervised social link inference on user trajectory data ([link](#))
Hyungho Byun, Yoonhyuk Choi, Chong-Kwon Kim
Information Sciences '23 (IF: 8.2)

- (C3) Review-Based Domain Disentanglement without Duplicate Users or Contexts for ... ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim
CIKM '22
- (C2) Finding Heterophilic Neighbors via Confidence-based Subgraph Matching for ... ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim
CIKM '22
- (J1) Dynamic graph convolutional networks with attention mechanism for rumor detection on social media ([link](#))
Jiho Choi, Taewook Ko, Yoonhyuk Choi, Hyungho Byun, Chong-Kwon Kim
PLOS ONE '21 (IF: 2.9)
.....
- (arXiv) Shifted Graph-Coupled Activation with Learned Edge Targets for Graph Neural Networks
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Jong-Wook Kim, Jiwoo Kang, Chong-Kwon Kim
Under Review (NeurIPS '26)
- (arXiv) Sparse Bayesian Message Passing under Structural Uncertainty
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Jiwoo Kang
Under Review (NeurIPS '26)
- (arXiv) Bias-aware Adaptive Loss for Sequential Recommendation
Yoonhyuk Choi, Jiho Choi, Chong-Kwon Kim
Under Review (CIKM '26)
- (arXiv) Adaptive Branch Specialization in Spectral–Spatial Graph Neural Networks for Certified Robustness
Yoonhyuk Choi, Chong-Kwon Kim
Under Review (TPAMI)
- (arXiv) Learning Sheaf Graph Neural Networks with PAC-Bayesian Spectral Guarantees
Yoonhyuk Choi, Yumin Lee, Chanran Kim, Chong-Kwon Kim
Under Review (TPAMI)
- (arXiv) Hierarchical and Uncertainty-Aware Graph Neural Networks for Heterophily and Robustness
Yoonhyuk Choi, Chanran Kim, Yumin Lee, Chong-Kwon Kim
Under Review (Pattern Recognition)
- (arXiv) Spectral-Radius and Uncertainty-Aware Blocking in Message-Passing Neural Networks
Yoonhyuk Choi, Jiho Choi, Yumin Lee, Chanran Kim, Chong-Kwon Kim
Under Review (TIST)

INVITED TALKS

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|---|-----------|
| • Co-Week Academy
From conventional AI to LLM (PyeongChang, South Korea) | Jun. 2026 |
| • DGIST Startup Camp
Connecting the dots (Seoul, South Korea) | Nov. 2025 |
| • N-EWN Partner Symposium
Identifying Potential Sites for Wetlands (St. Augustine, Florida) | May. 2024 |

RESEARCH PROJECTS

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|---|-----------|
| • Training Co-Evolving Human–AI to Lead On-Premise Generative Model in Financial Domain
<i>IITP Korea</i>
– Orchestration, Distillation, and Graph RAG | 2026-2029 |
|---|-----------|

- Development of Reliable, Personalized Clinical Decision-Support Technology**

NRF Korea

 - LLM and Graph RAG

2026-2029

- Enhancing Large Language Model with RAG**

Research project, Samsung SDS

 - Personalized RAG and knowledge graph construction on Knox Systems
 - Knowledge graph construction and RAG for chunk retrieval
 - Tiny-LLM (e.g., Llama 3B) distillation with huge-LLM (e.g., Llama 70B)

2025

- Can Large Language Model Improve the Text Convolution for Review-Based Recommendation?**

Research project, Emit Lab

 - Integrated the large language model (e.g., Llama 2, GPT-4) with the text convolution algorithm
 - Investigated whether applying summarization based on large language models first, instead of performing 2D convolution on the entire text, results in performance improvement
 - Used online shopping mall datasets like Amazon and Walmart

2024

- Selection Criteria and Assigned Weightage for Identifying Potential Locations Wetland**

Research project funded by NSF (in collaboration with Tulane University)

 - Suggested knowledge transfer between different regions and adaptive propagation between grids
 - Demonstrated the effectiveness of the framework through real-world scenario
 - Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets

2024

- Causal Discovery of Agricultural Mgmt and Reservoir Op. Induced Water Quality Change**

Research project funded by NSF (in collaboration with University of Arkansas)

 - Developed causal discovery algorithm for water quality improvement and reservoir management
 - Considered spatial and temporal variations and validated the causal learning ability
 - Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets

2023

- Tracking footprints with graph neural networks for the reduction of virus spread**

Coursework project, R&D in AI industry

 - Suggested spatial-temporal analysis for the next POI prediction to reduce virus spread
 - Selected as social contributing project
 - Used datasets are Coronamap of South Korea, Gowalla for POI prediction

2021 - 2022

- Personalized recommendation based on the user's purchasing histories and social network**

Industry project funded by Samsung Research

 - Introduced time series analysis of users' purchasing history for personalized advertising
 - Applied graph neural networks with binary recommendation techniques
 - Used customer datasets provided by Samsung Research

2020

- Next POI prediction based on user movements collected through large-scale sensors**

Research project funded by Samsung Electronics

 - Recommending the next place based on where students visited within Seoul National University
 - Developed energy-saving and effective multi-hop transmission technologies for sensor
 - Collected datasets by attaching special stickers to participants

2019

EXTRACURRICULAR ACTIVITIES

• Program Committee NeurIPS / ICLR / IJCAI / ICML / AAAI / KDD (Aug. track) / WSDM / TheWebConf	2026
• Program Committee ICLR / IJCAI / ICML / KDD (Feb. track) / CIKM / WSDM / TheWebConf	2025
• Program Committee LoG / CIKM / MM	2024
• Reviewer Journal of IEEE Multimedia / Plos one	2023
• Research Assistant (RA), Graduate Funded by Samsung Research	Mar. 2019 - Jun. 2021
• Teaching Assistant (TA) Topic: Social Network Analysis and Anomaly Detection (Advisor: Chong-Kwon Kim)	Mar. 2020 - Jun. 2020
• Research Assistant (RA), Undergrad Distributed Computing Lab (Supervisor: Jin-Suk Kim)	Jun. 2017 - Sep. 2017

AWARDS & GRANTS

• Best Ph.D. Dissertation Award Seoul National University	2023
• Overseas Short-term Training Scholarship Chonnam National University	2023
• BK21 Colloquium Graduate Student Fellowship Seoul National University	2023
• BK21 Star Student Researcher Fellowship Seoul National University	2023
• SIGIR Travel Awards For ACM Student Authors with Accepted Long Paper	2022
• BK21 Scholarship (Graduate) Seoul National University	2020
• Merit-based Scholarship (Undergrad) University of Seoul	2018 - 2019

SKILLS

- Languages:** Python, C, HTML/CSS
- Tools / Frameworks:** PyTorch, torch-geometric, Scikit-learn, Git, Django, AWS, LaTeX

REFERENCES

- ChongKwon Kim:** ckim@kentech.ac.kr
- TaeKyung Kwon:** tkkwon98@gmail.com
- U Kang:** ukang@snu.ac.kr